

PROJECT REPORT

Customer Segmentation

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1. Introduction

This report presents a customer segmentation analysis based on shopping data from customers in the U.S. market. The objective of the project is to understand customer shopping behaviour, identify discount opportunities, analyze spending patterns, and create customer segments for targeted marketing campaigns.

2. Problem Statement

The project focuses on the following business requirements:

- Identify product categories where discounts should be applied and provide proper reasoning.
- Analyze card/payment spending based on customer age.
- Explore the impact of seasons and locations on purchase amount.
- Perform customer clustering into three groups.
- Identify dominant factors and suggest targeted marketing campaigns based on shopping trends.

3. Dataset Overview

The dataset contains 3,900 customer shopping records. It includes customer demographic details, purchase behaviour, payment preferences, discount usage, review ratings, seasons, and locations.

- Key fields used: Age, Category, Purchase Amount, Season, Payment Method, Location, Review Rating, and Discount Applied.
- The dataset helps identify customer preferences, shopping trends, and business opportunities.
- The analysis supports data-driven decision-making for discount planning and customer targeting.

4. Tools and Methodology

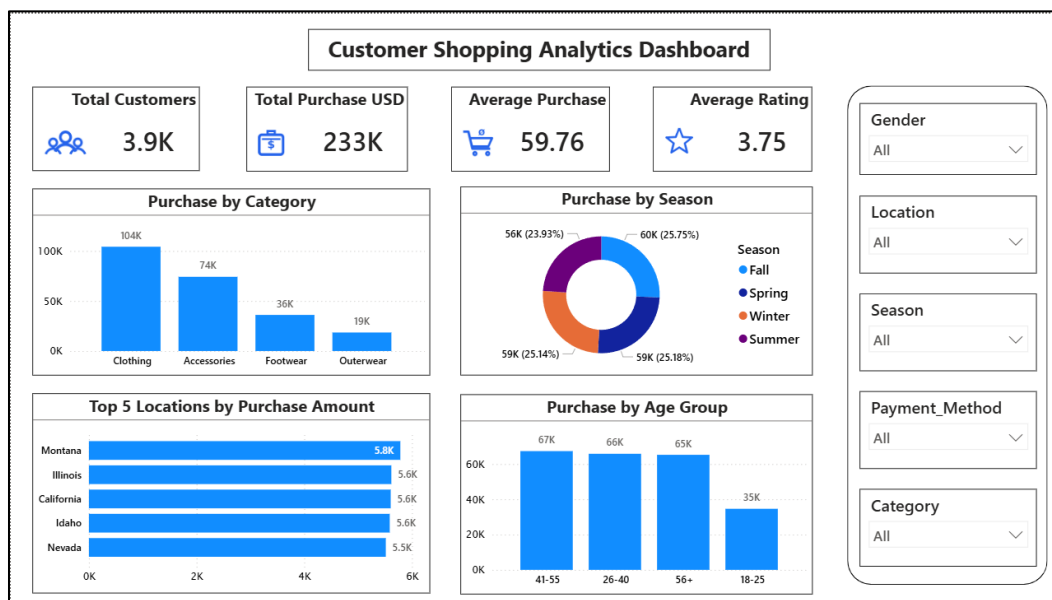
- Power BI Power Query was used for data import and cleaning.
- Power BI was used to create interactive dashboards and visual analysis.
- Orange Data Mining Tool was used to perform K-Means clustering.
- Customer segmentation was performed using Age and Purchase Amount as the main features.
- Clusters were mapped into business-friendly groups: High Value, Medium Value, and Low Value customers.

5. Dashboard Overview

The Power BI dashboard provides an overall summary of customer shopping behaviour. It includes KPI cards for Total Customers, Total Purchase Amount, Average Purchase, and Average Rating.

The dashboard also shows purchase trends by category, season, location, and age group. Interactive slicers were added for Gender, Location, Season, Payment Method, and Category to support filtered analysis.

Figure 1: Power BI Dashboard Overview



6. Discount Analysis

The discount analysis was performed to identify which product categories should receive discount-focused strategies.

- Clothing has the highest purchase amount and customer count among all product categories.
- Accessories is the second strongest category based on purchase amount and customer count.
- Discounted purchases are lower than non-discounted purchases across categories, which shows scope to improve the discount strategy.
- Discounts should be focused on Clothing and Accessories because these categories have a strong customer base and higher purchase contribution.

Business Recommendation: Clothing and Accessories should be prioritized for discount campaigns, seasonal offers, and repeat-purchase promotions to increase sales and customer retention.

7. Age, Payment and Location Analysis

This analysis was performed to understand how customer age groups, payment methods, seasons, and locations affect purchase amount.

- The 26–40 and 41–55 age groups show higher spending compared to younger customers.
- Credit Card and Venmo show strong spending across major age groups.
- Season-wise purchase amount is almost balanced, indicating consistent customer spending across seasons.
- Montana, Illinois, and California are among the top spending locations.

Business Recommendation: Marketing campaigns can be focused on high-spending age groups and top locations. Payment-based offers can be designed around Credit Card and Venmo users to increase conversion.

8. Customer Segmentation using K-Means

K-Means clustering was performed using Orange Data Mining Tool. The selected features for clustering were Age and Purchase Amount. These features were chosen because they directly represent customer demographic and spending behaviour.

- Customers were clustered into three groups using Age and Purchase Amount.
- The clusters were converted into business-friendly segments: High Value, Medium Value, and Low Value customers.
- High Value customers have the highest average purchase amount.
- Medium Value customers have the highest customer count and also contribute strongly to total purchase.
- Low Value customers have the lowest average purchase amount.
- Age and Purchase Amount were the dominant factors used for ML-based customer segmentation.

Figure 2: Customer Segmentation Dashboard

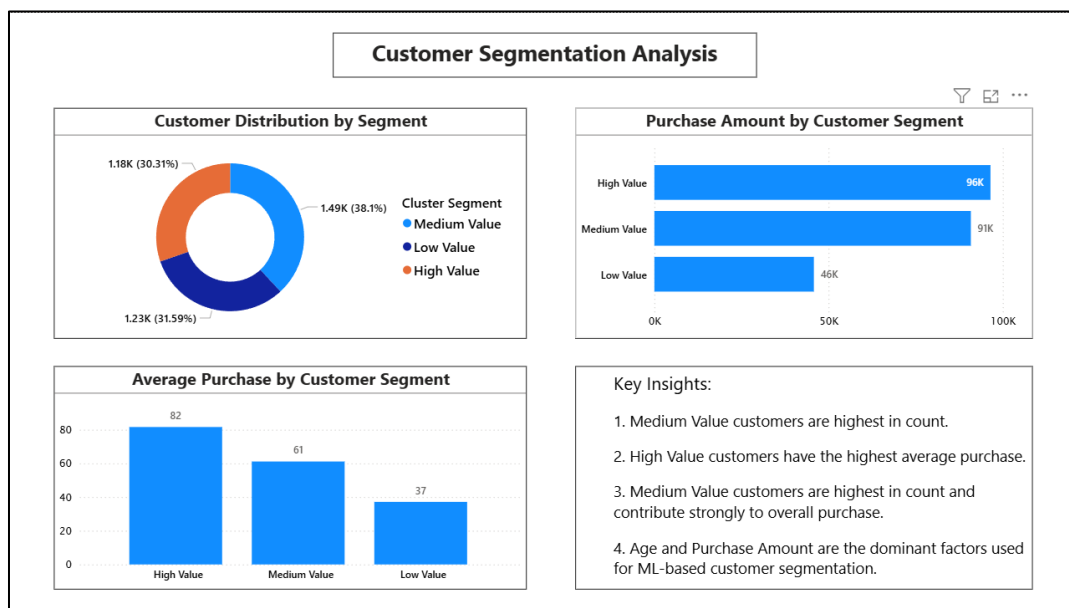
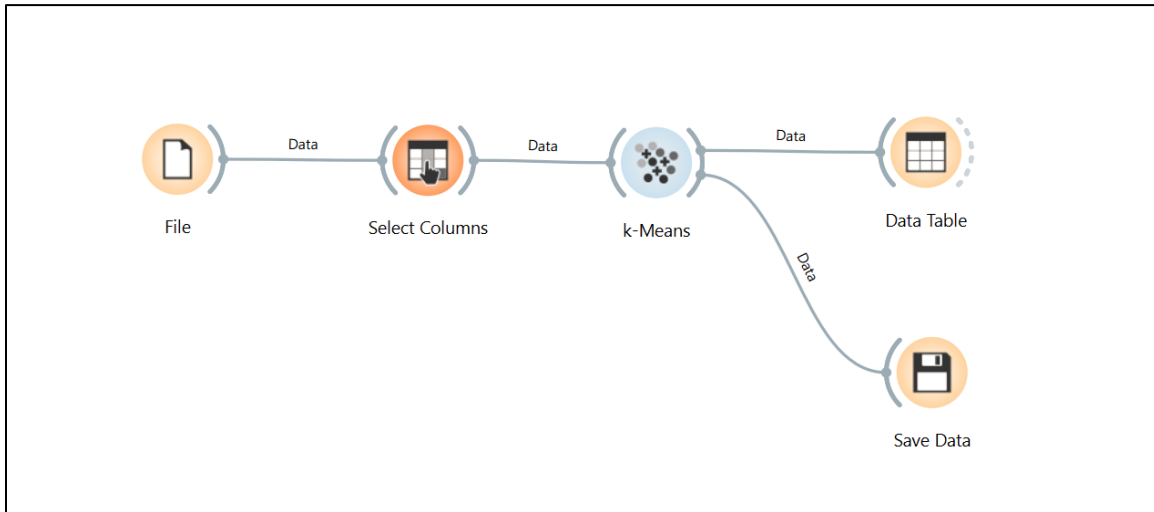


Figure 3: Orange K-Means Clustering Workflow

9. Cluster Mapping

The clusters generated by K-Means were mapped into business-friendly labels based on average Purchase Amount.

- C1 = High Value Customers
- C2 = Medium Value Customers
- C3 = Low Value Customers

The cluster with the highest average purchase amount was labelled as High Value, while the cluster with the lowest average purchase amount was labelled as Low Value. This mapping makes the machine learning output easier to understand for business decision-making.

10. Targeted Marketing Campaigns

Based on customer segmentation, targeted marketing campaigns were suggested for each customer group.

- High Value Customers: Provide loyalty rewards, VIP benefits, early access to new products, and premium product bundles.

- Medium Value Customers: Provide personalized discount coupons, combo offers, seasonal deals, and category-based recommendations to increase repeat purchases.
- Low Value Customers: Provide reactivation discounts, cashback offers, reminder campaigns, and attractive deals to increase purchase frequency.

These campaigns can help improve customer engagement, repeat purchases, and overall sales performance.

11. Conclusion

The project helped analyze customer shopping behaviour using Power BI and Orange Data Mining Tool. Clothing and Accessories were identified as the best categories for discount-focused strategies. Age, Purchase Amount, Payment Method, Season, and Location were used to understand customer spending patterns.

K-Means clustering was applied using Age and Purchase Amount to segment customers into High Value, Medium Value, and Low Value groups. Targeted marketing campaigns were suggested for each customer segment to improve engagement, repeat purchases, and sales.